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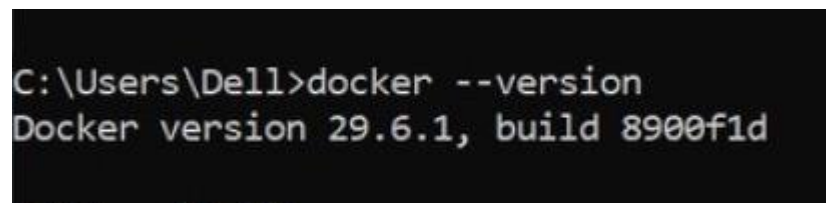
**Rollno: MCA2504104**

# Docker - Container Management

## Lab 2 : Docker Container Lifecycle and Management Using Nginx

### Step 1: Check Docker Installation

`docker --version`



```
C:\Users\Dell>docker --version
Docker version 29.6.1, build 8900f1d
```

### What is happening?

When you type:

`docker --version`

Docker checks whether the **Docker CLI (Command Line Interface)** is installed on your computer.

You

|

| `docker --version`

▼

Docker CLI

|

▼

Displays Version

Example Output

Docker version 28.2.2, build 123456. This command **does not contact Docker Hub or create anything**. It simply checks the Docker program installed on your machine.

# Step 2: Search for Nginx

docker search nginx

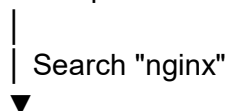
```
C:\Users\Dell>docker search nginx
```

| NAME                                   | DESCRIPTION                                                                                           | STARS | OFFICIAL |
|----------------------------------------|-------------------------------------------------------------------------------------------------------|-------|----------|
| nginx                                  | Official build of Nginx.                                                                              | 21337 | [OK]     |
| nginx/nginx-ingress                    | NGINX and NGINX Plus Ingress Controllers fo...                                                        | 121   |          |
| nginx/nginx-prometheus-exporter        | NGINX Prometheus Exporter for NGINX and NGINL...                                                      | 52    |          |
| nginx/nginx-ingress-operator           | NGINX Ingress Operator for NGINX and NGINX P...                                                       | 4     |          |
| nginx/nginxaas-loadbalancer-kubernetes |                                                                                                       | 1     |          |
| bitnamicharts/nginx                    | Bitnami Helm chart for NGINX Open Source                                                              | 4     |          |
| ubuntu/nginx                           | Nginx, a high-performance reverse proxy & we...                                                       | 141   |          |
| kasmweb/nginx                          | An Nginx image based off nginx:alpine and in...                                                       | 9     |          |
| rancher/nginx                          |                                                                                                       | 4     |          |
| linuxserver/nginx                      | An Nginx container, brought to you by LinuxS...                                                       | 236   |          |
| dtagdevsec/nginx                       | T-Pot Nginx                                                                                           | 0     |          |
| paketobuildpacks/nginx                 |                                                                                                       | 0     |          |
| vmware/nginx                           |                                                                                                       | 3     |          |
| gluufederation/nginx                   | A customized NGINX image containing a consu...                                                        | 1     |          |
| cleanstart/nginx                       | Secure by Design, Built for Speed, Hardened ...                                                       | 0     |          |
| antrea/nginx                           | Nginx server used for Antrea e2e testing                                                              | 0     |          |
| activestate/nginx                      | ActiveState's customizable, low-to-no vulner...                                                       | 0     |          |
| intel/nginx                            |                                                                                                       | 0     |          |
| docksal/nginx                          | Nginx service image for Docksal                                                                       | 1     |          |
| geokrety/nginx                         | Our customized nginx image                                                                            | 0     |          |
| circleci/nginx                         | This image is for internal use                                                                        | 2     |          |
| ilios/nginx                            | Nginx customized to run Ilios along with the...                                                       | 0     |          |
| corpusops/nginx                        | <a href="https://github.com/corpusops/docker-images/">https://github.com/corpusops/docker-images/</a> | 1     |          |
| wayofdev/nginx                         | Nginx Docker image for PHP development. SSL-...                                                       | 0     |          |
| dockette/nginx                         | Nginx SSL / HSTS / HTTP2                                                                              | 3     |          |

## What happens?

Docker connects to **Docker Hub**, which is the official online repository for Docker images.

Your Computer



Docker Hub



Returns Matching Images

Example Output

| NAME           | DESCRIPTION             |
|----------------|-------------------------|
| nginx          | Official build of Nginx |
| nginxinc/nginx |                         |
| bitnami/nginx  |                         |

## Why?

Docker images are stored online. Before downloading one, you can search for available images.

Think of Docker Hub like the **Google Play Store** or **Apple App Store**, but instead of apps, it stores Docker images.

## Step 3: Download Nginx

docker pull nginx

```
C:\Users\Dell>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
45ef33eb91c8: Pull complete
6a24a6ebcf30: Pull complete
0b19be717994: Pull complete
8d2092eabbc5: Pull complete
3b9f3509a826: Pull complete
062e450697fa: Pull complete
39694cde734e: Pull complete
cd64a2895680: Download complete
29456554d205: Download complete
Digest: sha256:c90e4ea5905ee69515d84cd5520e11d204df06cce71a312666cebbbc4b9b68267
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview nginx
```

### What happens?

Docker downloads the **Nginx image** from Docker Hub to your computer.

Docker Hub



Download Image



Your Computer



Stored in Docker Images

After downloading:

docker images

shows

| REPOSITORY | TAG    |
|------------|--------|
| nginx      | latest |

## Important

At this point:

✗ No container exists.

Only the **image** exists.

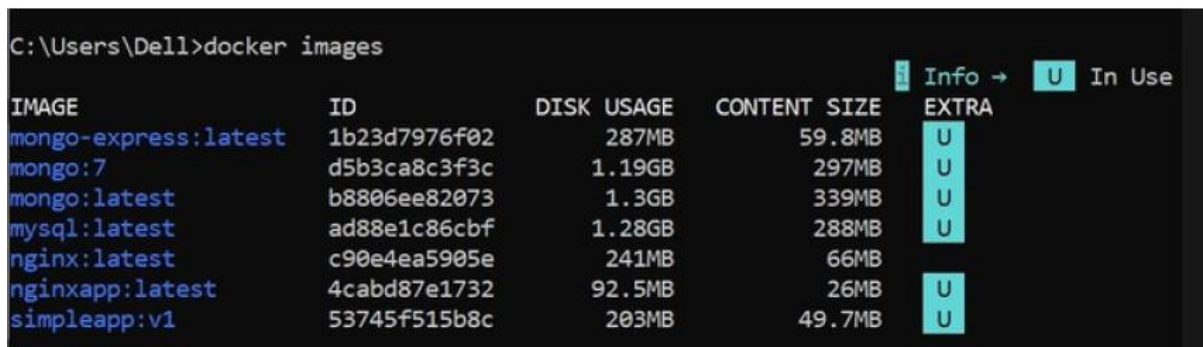
Think of it like:

Image = Software Installer

Container = Installed & Running Software

## Step 4: View Images

docker images



| IMAGE                | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|----------------------|--------------|------------|--------------|-------|
| mongo-express:latest | 1b23d7976f02 | 287MB      | 59.8MB       | U     |
| mongo:7              | d5b3ca8c3f3c | 1.19GB     | 297MB        | U     |
| mongo:latest         | b8806ee82073 | 1.3GB      | 339MB        | U     |
| mysql:latest         | ad88e1c86cbf | 1.28GB     | 288MB        | U     |
| nginx:latest         | c90e4ea5905e | 241MB      | 66MB         | U     |
| nginxapp:latest      | 4cabd87e1732 | 92.5MB     | 26MB         | U     |
| simpleapp:v1         | 53745f515b8c | 203MB      | 49.7MB       | U     |

## What happens?

Docker shows all downloaded images.

Example

IMAGE ID

nginx

ubuntu

mysql

Nothing is running yet.

You only have installers.

# Step 5: Run Nginx

`docker run nginx`

This is one of the most important Docker commands.

```
C:\Users\Dell>docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 08:59:22 [notice] 1#1: using the "epoll" event method
2026/07/14 08:59:22 [notice] 1#1: nginx/1.31.2
2026/07/14 08:59:22 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 08:59:22 [notice] 1#1: OS: Linux 6.6.87.1-microsoft-standard-WSL2
2026/07/14 08:59:22 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 08:59:22 [notice] 1#1: start worker processes
2026/07/14 08:59:22 [notice] 1#1: start worker process 29
2026/07/14 08:59:22 [notice] 1#1: start worker process 30
2026/07/14 08:59:22 [notice] 1#1: start worker process 31
2026/07/14 08:59:22 [notice] 1#1: start worker process 32
```

## What happens internally?

Docker performs several actions:

### Step A

Checks whether the image exists.

Is nginx image available?

YES

If not:

Docker automatically performs:

`docker pull nginx`

## Step B

Creates a **new container** from the image.

Image



New Container

The image remains unchanged.

Containers are copies of the image.

## Step C

Starts the Nginx server.

Container



Nginx Starts

Now Nginx is running inside the container.

## Step D

Your terminal becomes attached to the running container.

Terminal



Container Output

Stopping the program using

Ctrl + C

stops the container.

## Step 6: Detached Mode

`docker run -d nginx`

What changed?

`-d`

means

Detached Mode

Instead of connecting your terminal to the container:

Terminal

↓

Gets Prompt Back

The container continues running in the background.

Docker returns

`9f3ab2c89c8d...`

This is the **Container ID**.

## Step 7: Name the Container

`docker run -d --name mynginx nginx`

Without a name:

Docker generates random names.

Example

`adoring_pike`

`crazy_turing`

`happy_einstein`

Using

```
--name mynginx
```

makes management easier.

Instead of

```
docker stop 98ab123...
```

you simply type

```
docker stop mynginx
```

## Step 8: Publish a Port

```
docker run -d --name webserver -p 8080:80 nginx
```

This is the most important networking concept.

### Inside the container

Nginx always listens on

Port 80

Container

Nginx

↓

80

But your computer cannot automatically access that internal port.

So Docker creates a mapping.

Computer

8080

↓



Container

80

Meaning

localhost:8080

↓

Container Port 80

Opening

<http://localhost:8080>

actually forwards the request to

Container

↓

Nginx

↓

Port 80

## Step 9: List Running Containers

`docker ps`

Shows

Container ID

Image

Status

Ports

Name

Example

mynginx

Up 5 minutes

0.0.0.0:8080->80

Meaning

Host Port

8080

↓

Container Port

80

## Step 10: Stop Container

```
docker stop mynginx
```

Docker sends a signal to Nginx.

Please Stop Gracefully

Nginx saves its state and exits cleanly.

Container Status

Running

↓

Exited

Notice:

The **container is not deleted**.

It is only stopped.

## Step 11: Start Container Again

```
docker start mynginx
```

Docker does **not create a new container**.

It simply starts the **existing** one.

Stopped Container

↓

Running Container

## Step 12: Restart

```
docker restart mynginx
```

Equivalent to

```
docker stop mynginx
```

```
docker start mynginx
```

## Step 13: View Logs

```
docker logs mynginx
```

Every application writes messages.

Example

Nginx Started

Request Received

Response Sent

Docker stores these logs.

Useful for debugging.

## Step 14: Enter the Container

```
docker exec -it mynginx bash
```

This command opens a shell **inside the running container**.

Your Computer

↓

Docker

↓

Container

↓

Linux Bash

Now commands like

```
pwd
```

```
ls
```

```
cd /usr/share/nginx/html
```

are executed **inside the container**, not on your Windows machine.

This is very useful for troubleshooting or inspecting files.

## Step 15: Remove the Container

```
docker rm mynginx
```

Docker deletes the **container only**.

Container

✗ Deleted

The image is still available.

You can create a new container anytime using:

```
docker run nginx
```

## Step 16: Remove the Image

```
docker rmi nginx
```

Now Docker removes the **downloaded image** from your local system.

Image

✗ Deleted

If any container still depends on that image, Docker will refuse to remove it until those containers are deleted.